

Q1

Employee department listing

List the names of all employees along with their department name and location. Include employees who have not been

assigned to any department.

```
select e.name, d.dept_name, d.location
from employees as e left join departments as d
on e.dept_id=d.dept_id;
```

Q2

Above-average earners

Find the names and salaries of all employees who earn more than the average salary of the entire company.

```
select name, salary from employees
where salary > (select avg(salary) from employees);
```

Q3 Outearning the Finance team

Retrieve the names of employees whose salary is greater than the maximum salary of every individual employee in the

Finance department (dept_id = 30).

```
select name, salary from employees
where salary > (select max(salary) from employees where dept_id=30)
```

Q4

For each employee, find their name, salary, and the average salary of their own department. Display only employees

who earn above their department's average salary using subquery.

```
select e.name, e.salary,
(select avg(e2.salary) from employees as e2 where e2.dept_id = e.dept_id) as dept_avg_salary
from employees as e
where e.salary > (select avg(e2.salary) from employees as e2 where e2.dept_id = e.dept_id);
```

Q6

For each completed order, display the order ID, customer ID, order date, amount, and a 3-order rolling average of the

amount (current row and the 2 preceding rows) ordered by order date. Round the rolling average to 2 decimal places.

```
select order_id, customer_id, order_date, amount,  
round( avg(amount) over(order by order_date  
rows between 2 preceding and current row), 2) as rolling_avg_3  
from orders where status = 'completed';
```

Q7

Products that have been sold

Find the names and categories of all products that have been sold at least once. Use the EXISTS keyword.

```
select p.product_name, p.category from products as p  
where exists (select 1 from sales where s.product_id = p.product_id);
```

Q8

GROUP BY + HAVING — COMMON INTERVIEW

Departments with more than 2 employees

Find each department that has more than 2 employees, along with the number of employees and the highest salary in

that department.

```
select d.dept_name, count(e.emp_id) as employee_count, max(e.salary) as max_salary  
from departments as d join employees as e  
on d.dept_id = e.dept_id  
group by d.dept_id, d.dept_name  
having count(e.emp_id) > 2;
```

Q9 Explain Normalization

normalization in sql is the process of organizing data in tables to reduce redundancy and improve data integrity.

types of normalization are:

1. 1NF= IN first normal form each coloumn should contain atomic values
2. 2NF= Second normal form must be in 1nf no partial dependency on part of a composite key
3. 3NF= Third normal form must be in 2nf no transitive dependency

Q10 Explain CTE Vs Views

CTE= A cte (common table expression) is a temporary result set that exists only for the duration of a single sql query.

View= A view is a saved sql query that can be reused later